YAKEEN

NEET 2.0 2027

Physics | Lecture - 9

Motion in a Straight Line

By- Manish Raj (MR Sir)



H/w goal

→ Sangh Assignment - 4 of 1-D

Todays Goal

PhD on graph:

Ex- x/t

v/t

v/x

a/t



Question

$72 \times \frac{5}{18} = 20$

A bus moving along a straight highway with speed of 72 km/h is brought to halt within 4s after applying the brakes. The distance travelled by the bus during this time (Assume the retardation is uniform) is _____ m. **[JEE Main 2024]**

$S = \frac{u^2}{2a}$

$S = \left(\frac{u+v}{2}\right)T$

$= \left(\frac{20+0}{2}\right) \times 4 = 40\text{m}$

(9PY

(a) If $u_i = 25\ m/s \longrightarrow S$ (stopping distn) $= 8\ Km$.

If $u_i = 75\ m/s \longrightarrow S$ (") $= ??$

$$S \propto u^2$$

$$\frac{8\ km}{S} = \left(\frac{\cancel{25}}{\cancel{75}_{3}}\right)^2 = \frac{1}{9}$$

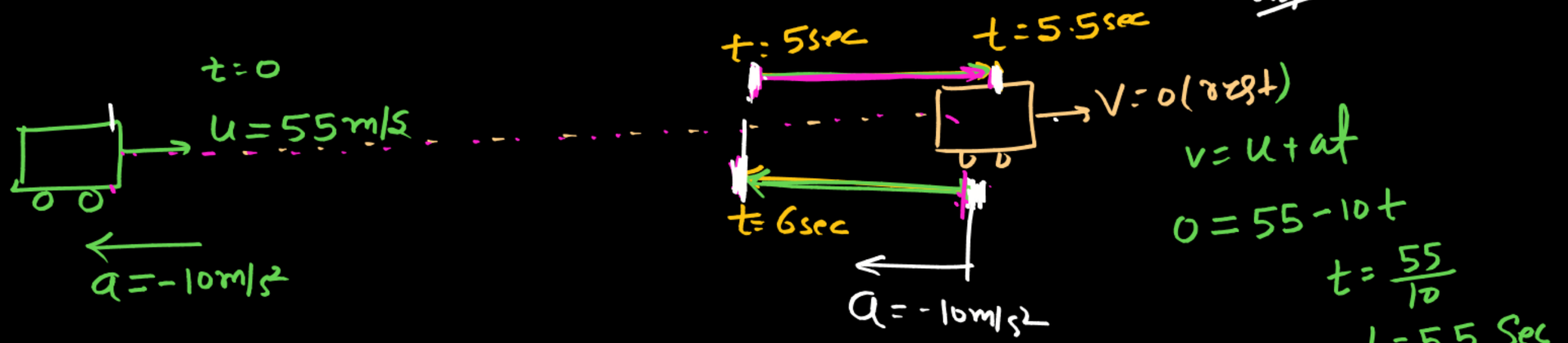
$$S = 72\ km$$

Question

copy

CAR is moving with initial speed 55 m/s and retardation $-10\ m/s^2$ find distance in 6th sec.

displ^n



$v = u + at$

$0 = 55 - 10t$

$t = \frac{55}{10}$

$t = 5.5$ Sec

$S_{displ^m \text{ in } 6^{th} sec} = 0$ ✓

$= u + \frac{a}{2}(2n-1)$

$= 55 - \frac{10}{2}(2\times6-1)$ ✓

$= 55 - 5 \times 11$

$= 0\ m$

$S = \frac{1}{2}(-10)\left(\frac{1}{2}\right)^2$

$= \frac{-5}{4} = 1.25\ m$

total displ^n

$= 1.25 + 1.25$

$= 2.5\ m$ ✓

Question

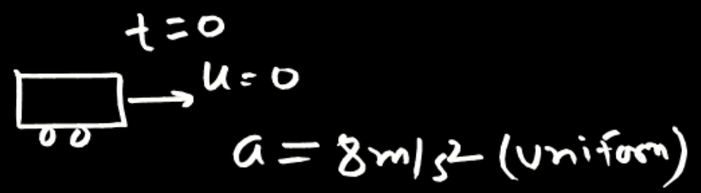
done

Object starts his motion from rest and constant acceleration 8 m/s^2 then find displacement in 5th sec.

Question

copy

Object starts his motion from rest and constant acceleration 8 m/s^2 then find displacement between $t = 2$ sec to $t = 8$ sec?



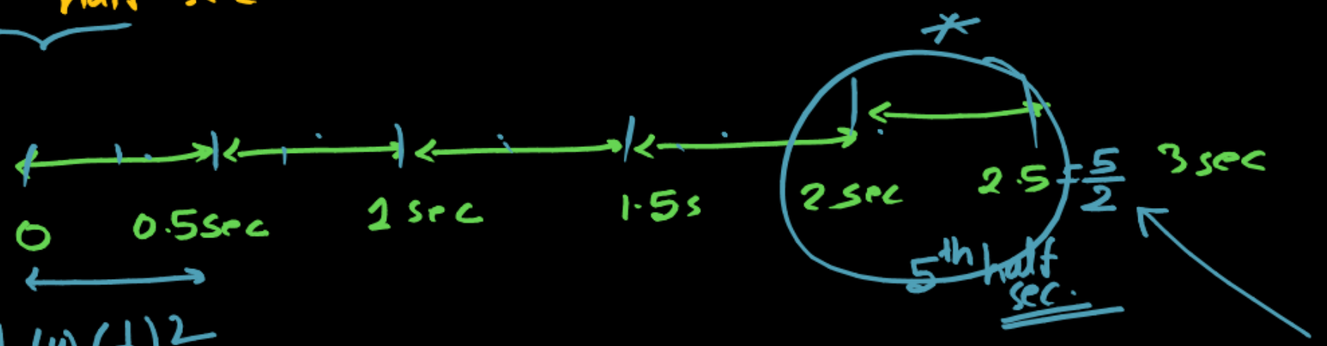
$$S_{t=2s} = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}\cancel{8}^{4}(2)^2 = 16m$$

$$S_{t=8} = 0 + \frac{1}{\cancel{2}}\cancel{8}^{4}(8)^2 = 4 \times 64 = 256m$$

$$\text{disp}^m \text{ B/ω 2 to 8s} = 256 - 16$$

$$= 240m$$

(8) object starts motn from rest & costn accn $4\,m/s^2$ find dispm. in 5th half sec.



$$S = \frac{1}{2}(4)\left(\frac{1}{2}\right)^2$$

$$= 0.5m$$

$$S_{5^{th}\ half\ sec} = S_{2.5sec} - S_{2sec}$$

$$= \frac{1}{2}(4)\left(\frac{5}{2}\right)^2 - \frac{1}{2}(4)(2)^2$$

$$= \frac{2 \times 25}{4_2} - 2 \times 4$$

$$= 12.5 - 8$$

$$= 4.5m$$

Galileo refer vale

(8) object starts his motion with 10m/s and cost^n acc^n moves 40m in 1st 2-sec then find disp^m in 2sec - to 4sec

copy

Sol^n

radio not valid

Ans → 120m

(a) yes (18%)

(b) NO

intial velocity is not equal zero.

$u = 10m/s$ $a = ?$

$S = 40m$ $t = 2sec$

$S = ut + \frac{1}{2}at^2$

$40 = 10 \times 2 + \frac{1}{2}a(2)^2$

$20 = \frac{1}{2}a(4)$

$a = 10m/s^2$

$S_{t=4s} = ut + \frac{1}{2}at^2$

$= 10 \times 4 + \frac{1}{2} \times 10 (4)^2$

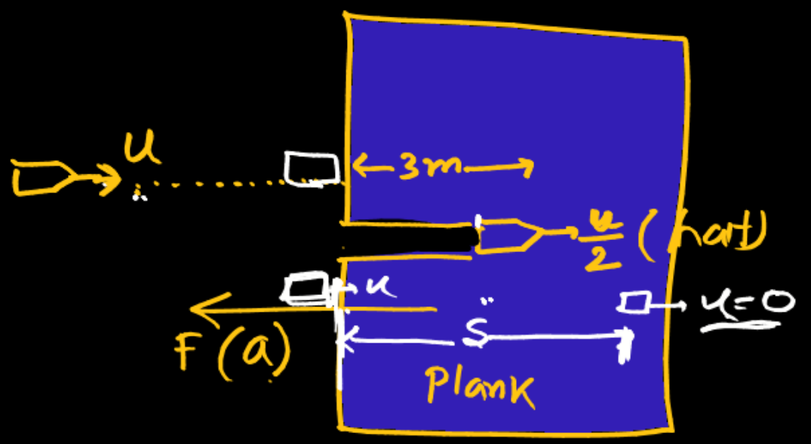
$= 40 + 80$

$= 120m$

$S_{2 \text{ to } 4s} = 120 - 40$

$= 80m$

Copy ⑥



if Uniform retardation Produce by plank then find further distance after which bullet will stop.

Soln use 3rd eqn of motion ✓

$$\left(\frac{u}{2}\right)^2 - u^2 = 2a(3) \quad - (1)$$

$$0 - u^2 = 2a(s) \quad - (ii)$$

$$\frac{-3u^2}{4(-u^2)} = \frac{2a \times 3}{2a\,s}$$

$$3(s) = 4(3)$$

$$s = 4$$

$$s\ (\text{Total in plank}) = 4$$

$$\text{further dis} = 4 - 3$$

$$= 1\text{m}$$

(1) If velocity of object $V = \sqrt{49 - 8x}$ then find acc^n & intial velocity of object.

Soln $V = \sqrt{49 - 8x}$ ⟶ find $a = ??$

$a = v\frac{dv}{dx}$

acc^n = cost^n

3rd eqn of mot^n.

$v^2 = 7^2 - 8x$ $[v^2 = u^2 + 2ax]$

compare

$u^2 = 7^2$

$u = 7 m/s$

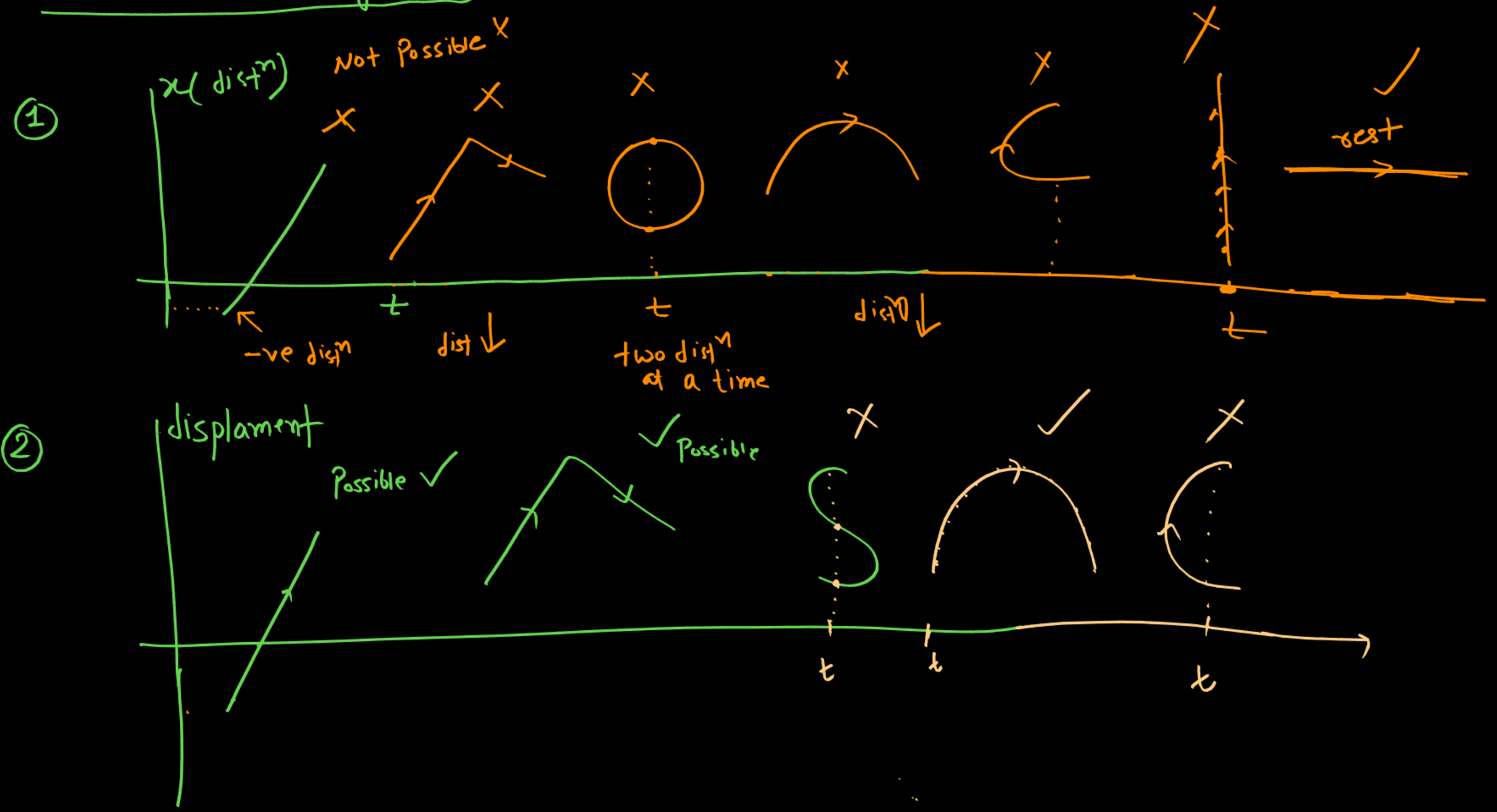
$-8x = 2ax$

$-8 = 2a$

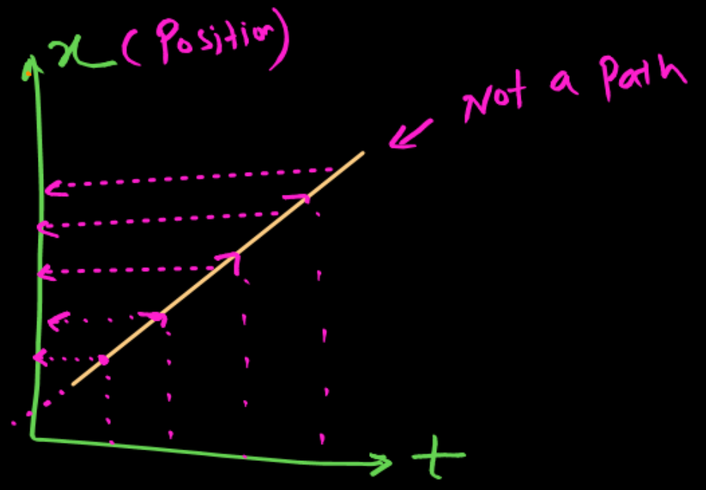
$a = -4 m/s^2$

Let's starts graph

distance - time graph



Position - time graph



① Position (x) ⟶ ↑

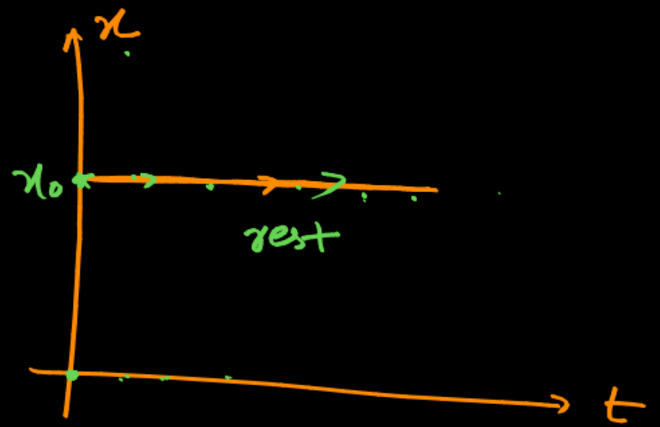
② time (t) ⟶ ↑

③ $\text{Slope} = \frac{dx}{dt} = \tan\theta$

$\uparrow_{\text{Const}^n} = \text{velocity}$

④ $\text{Area} = \int x\,dt = \text{Not a } \underline{\text{P.O.}}$

(3)



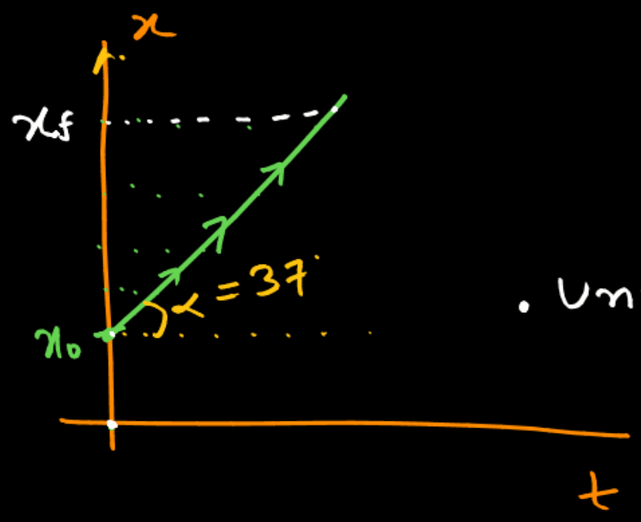
time is ↑ but

- x (Position) = x_0 (fixed) $cost^n$.
- velocity = zero

$\frac{dx}{dt}$ (slope) = velocity = 0

{ • $acc^n = 0$ }

(4)



• Uniform motion

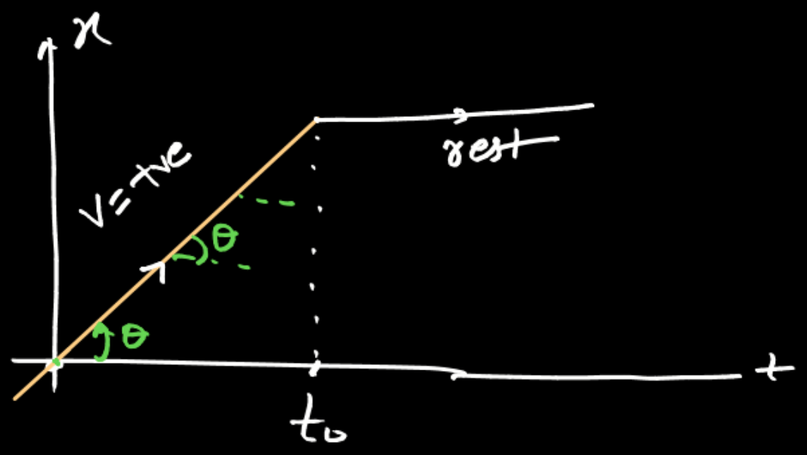
at $t = 0$ Position = x_0 ⇒ x ↑ continously with t

Velocity = $\frac{dx}{dt}$ (slope) = $cost^n = \tan\alpha$

{ $acc^n = 0$ } $V = \tan 37° = \frac{3}{4}$ m/s

$disp^m = x_f - x_0$

(5)



initial Position = 0 at t = 0 → x ↑ & cont^n after t_0

Initial velocity (t=0) = +ve ($\tan\alpha$) cont^n upto t_0; after t_0 velocity is zero.

Uniform Motion upto t_0

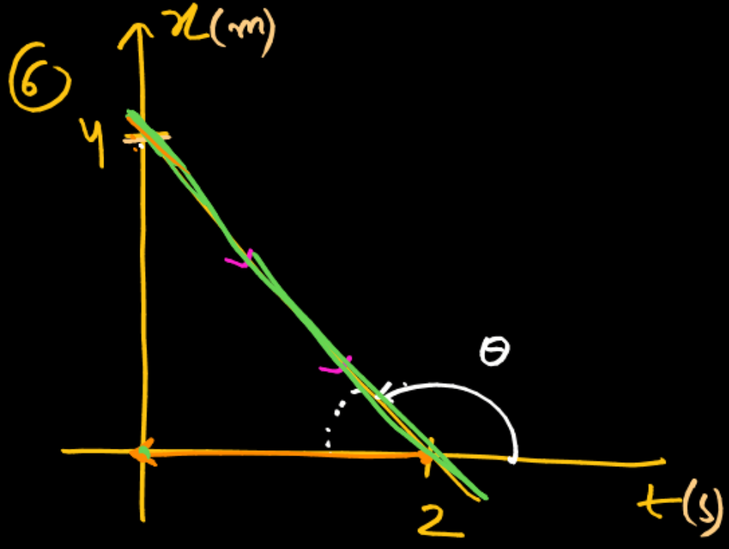
$\boxed{a=0}$

Avg velocity

$= \frac{0-4}{2}$

$= -2\ m/s$

(6)



at t=0 $x_i = 4m$

& x (Position) ↓ with time

Velocity = -ve & cont^n

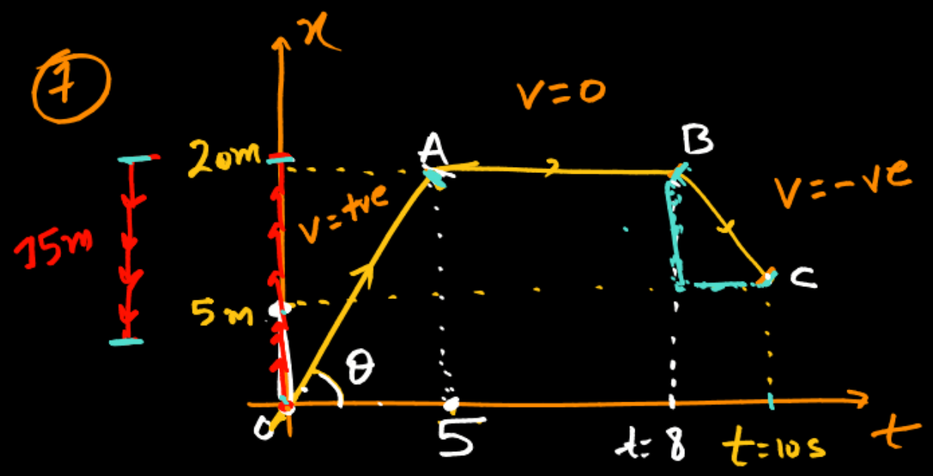
$= -\frac{4}{2} = -2\ m/s$

cont^n.

acc^n = 0

disp^n = 0 - 4 = -4m

(7)



Soln

OA → V = +ve & costⁿ = $\frac{20}{5}$ = +4m/s

at t=0 V → 4m/s

t=1s V → 4m/s

t=1.8se

t=1.9se

t=2.8s

t=3.9

t=4.9

t=4.995s

} V=4m/s

AB → rest

t=5.1sec V=0

t=6se V=0

BC → V=-ve costⁿ.

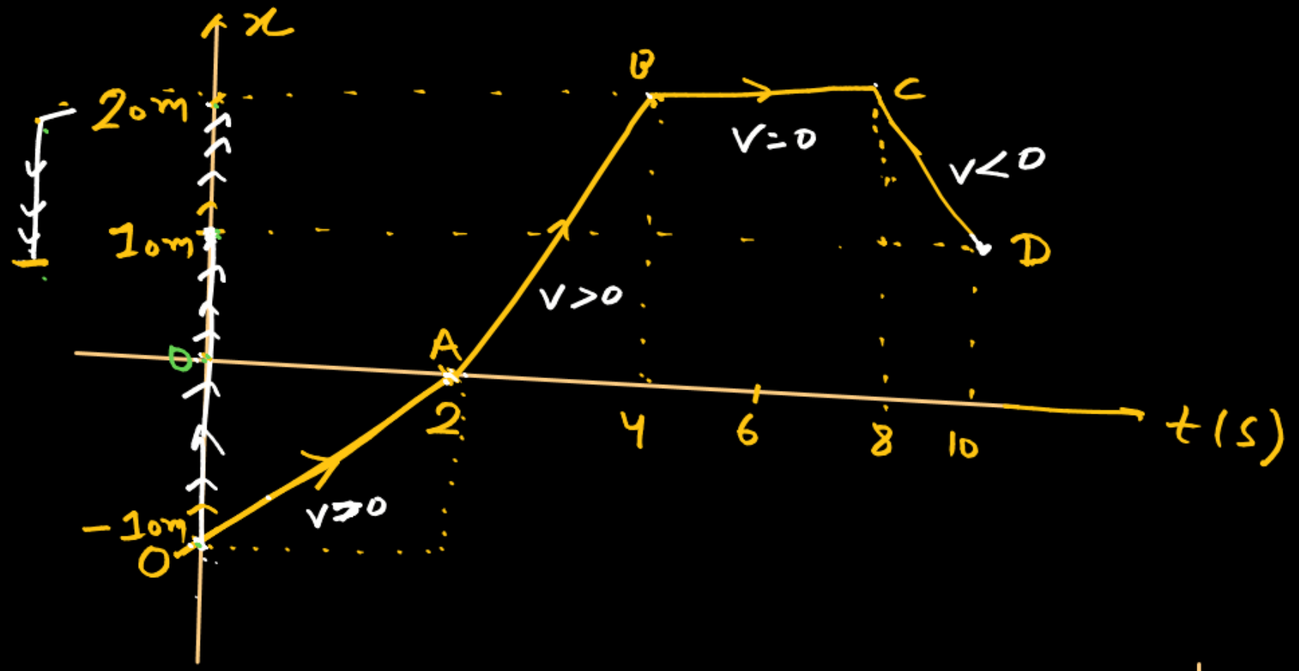
V = $\frac{-15}{2}$ = -7.5m/s

Avg velocity [0-10]sec = $\frac{\text{displ}^m}{\text{time}}$ = $\frac{5-0}{10}$ = $\frac{1}{2}$ m/s

Avg speed = $\frac{\text{total dist}}{\text{total time}}$ = $\frac{35}{10}$ = 3.5m/s

Avg accⁿ = $\frac{V_f - V_i}{\Delta t}$ = $\frac{-7.5-4}{10}$ = $\frac{-11.5}{10}$ m/s²

(8)



Soln

(i) OA ⟶ V = +ve = $+\frac{10}{2}$ = 5m/s

(ii) AB ⟶ V = +ve = $\frac{20}{2}$ = 10m/s

(iii) BC ⟶ rest

(iv) CD ⟶ V = -ve = $-\frac{10}{2}$ = -5m/s

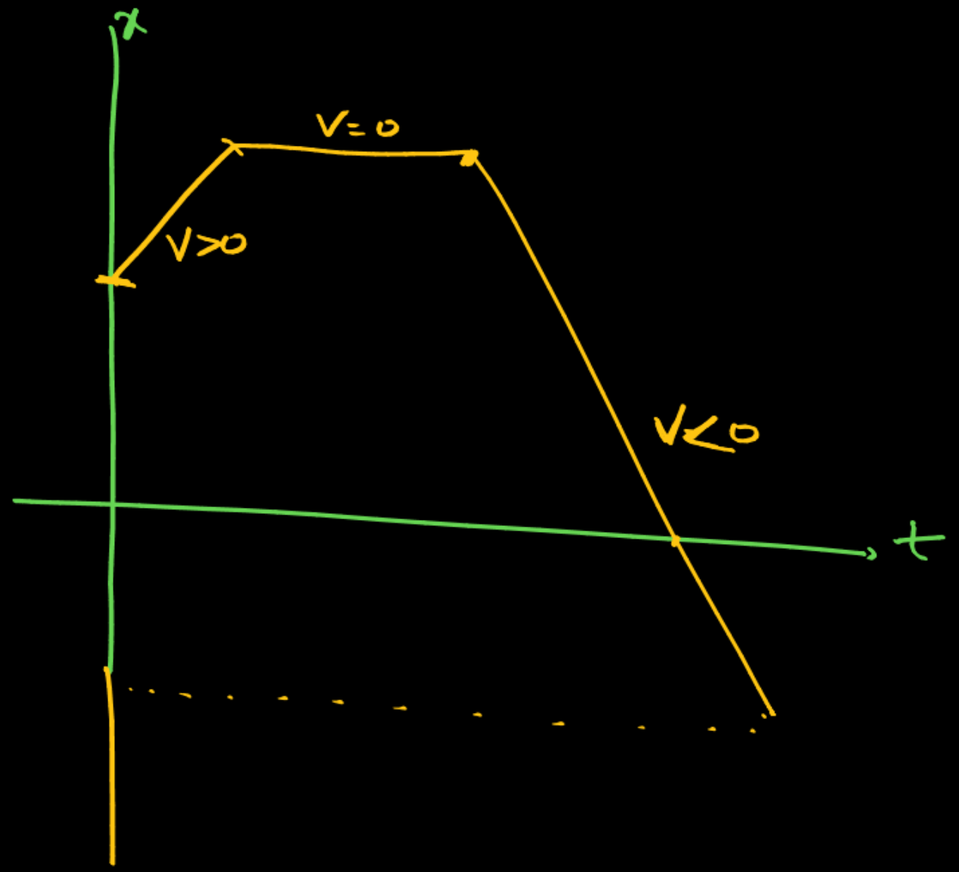
Avg velocity (0–10s) = $\frac{20m}{10}$ = 2m/s

Avg speed = $\frac{40}{10}$ = 4m/s

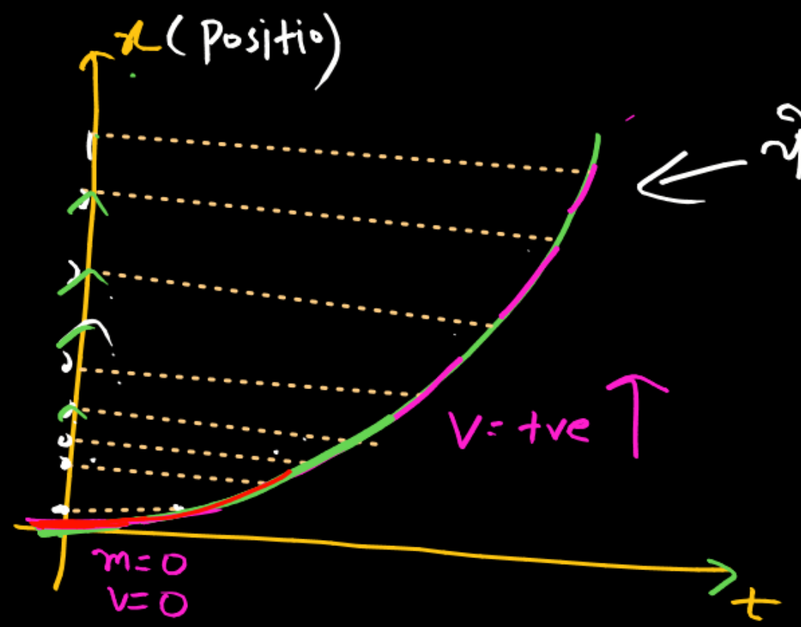
Avg acc^{n} in Compt^{n} = $\frac{\vec{V_f} - \vec{V_i}}{\Delta t}$

$= \frac{-5-5}{10} = -1 m/s^2$

(5)



(To)



← यह रास्ता नहीं है

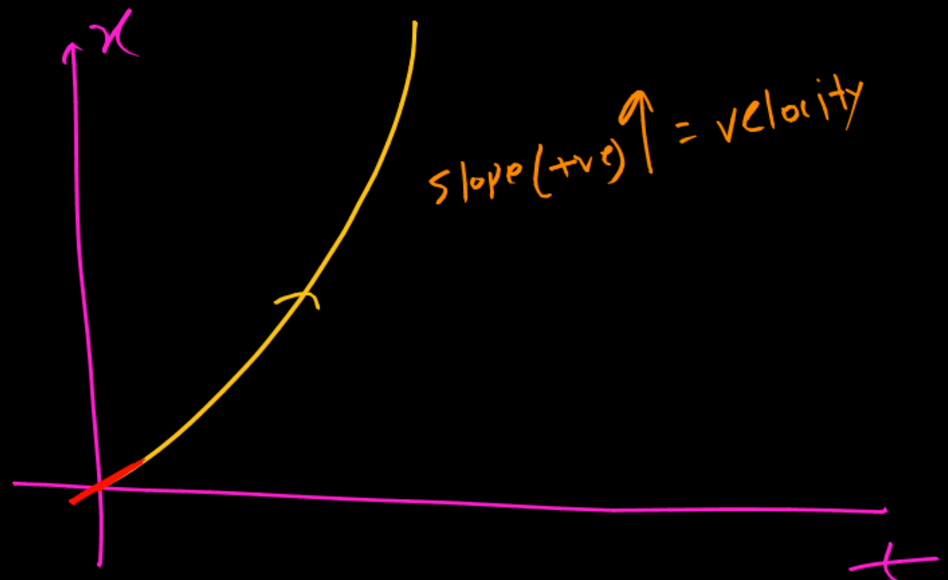
⇒ Position $x_i = 0$ at $t = 0$ & then $x \uparrow$ continously.

⇒ at $t = 0$ v = slope $\left(\frac{dx}{dt}\right) = 0$ then velocity = +ve ↑
speed (↑)

⇒ $acc^n \neq 0$
+ve acc^n

imp
If u & a have same sign (dir) speed ↑
If u & a opposite sign then speed ↓

copy (11)

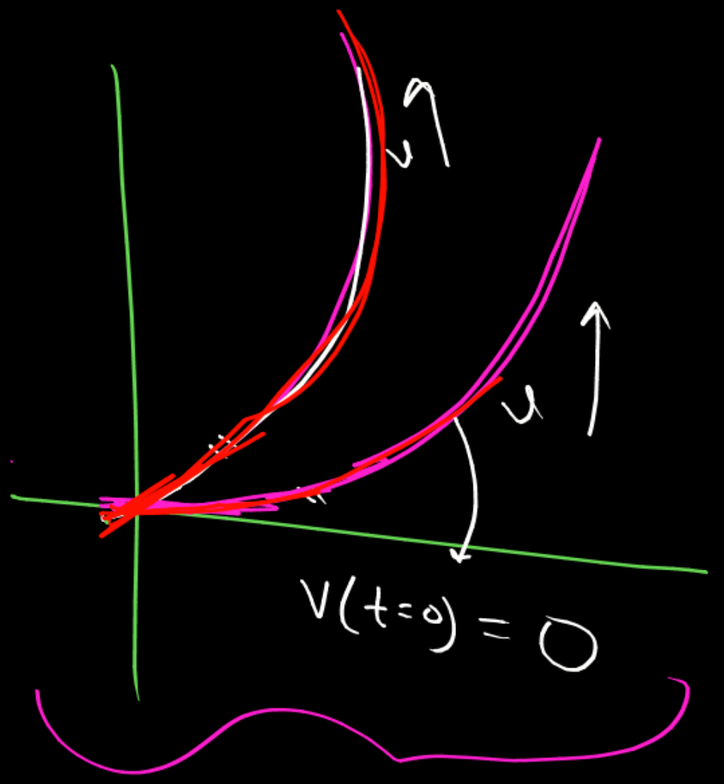


Soln

at $t=0 \rightarrow u = +ve$

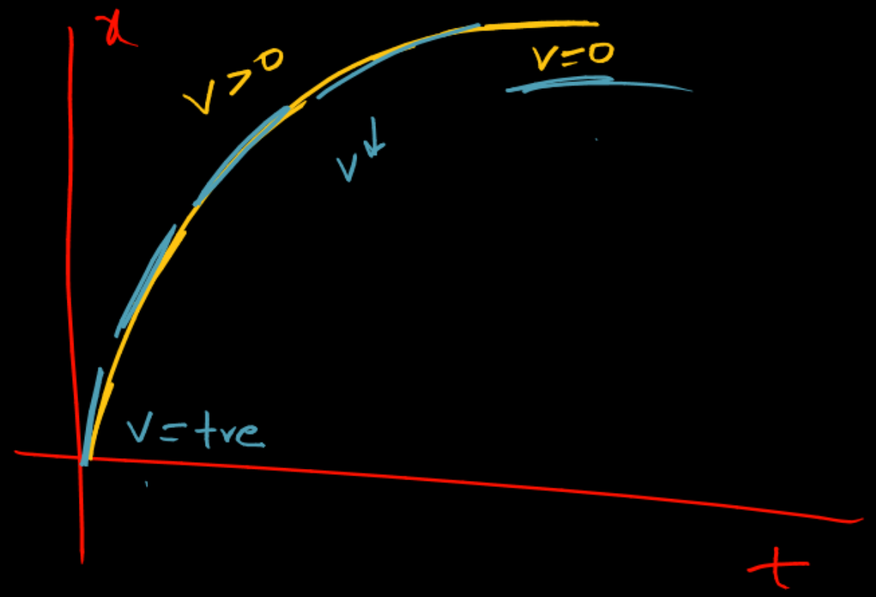
with time $u = +ve \uparrow$

$a = +ve$



No copy

(12)



Soln

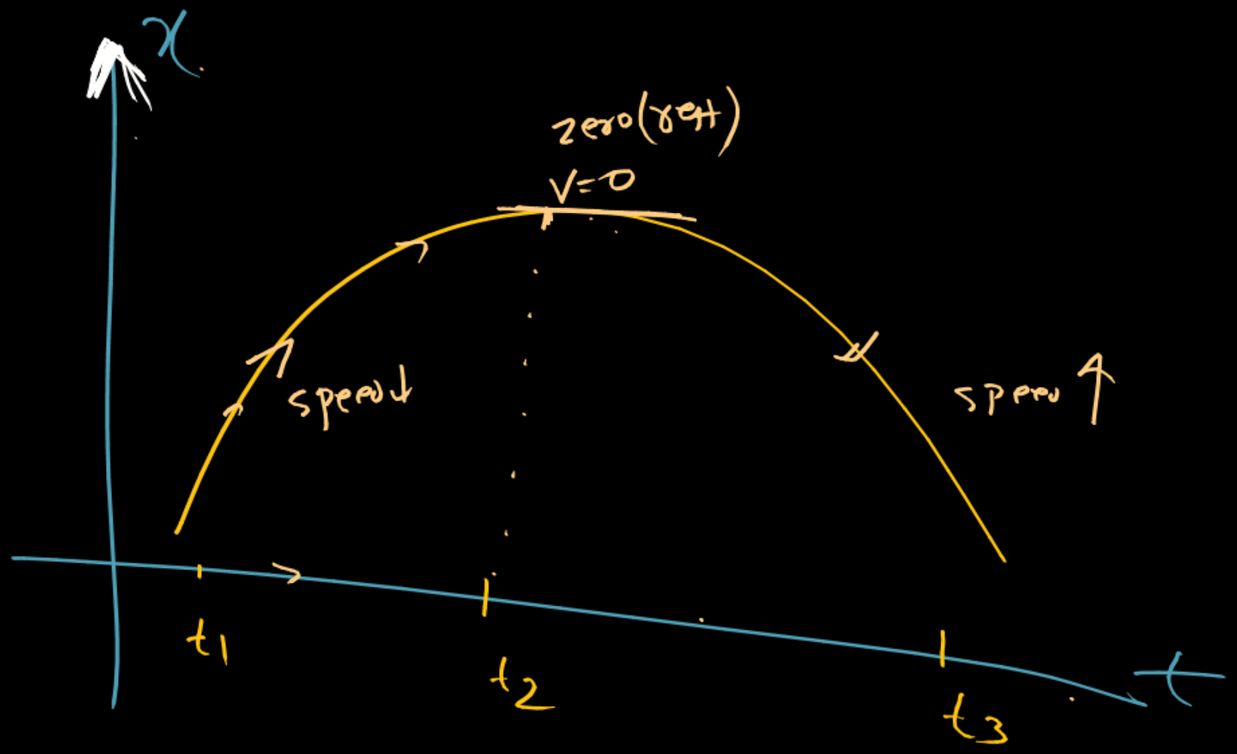
at $t=0$ $\underline{v = +ve}$ (large)

speed ↓

$v \to +ve$ & ↓

acc^n = (a) +ve ✗
(b) -ve ✓

13



Sol^n

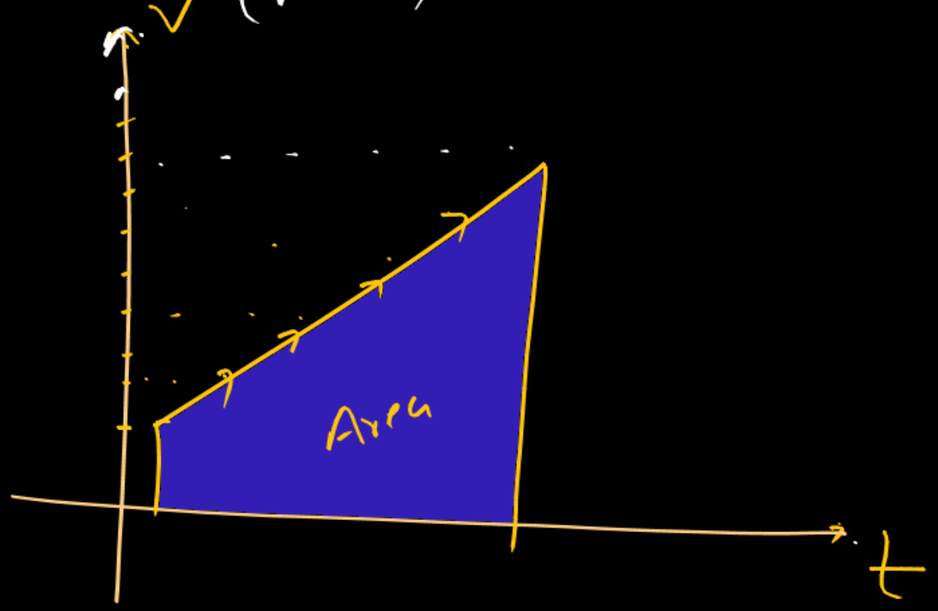
(speed ↑)

on which time interval acc^n & velocity is Parallel → t_2 to t_3

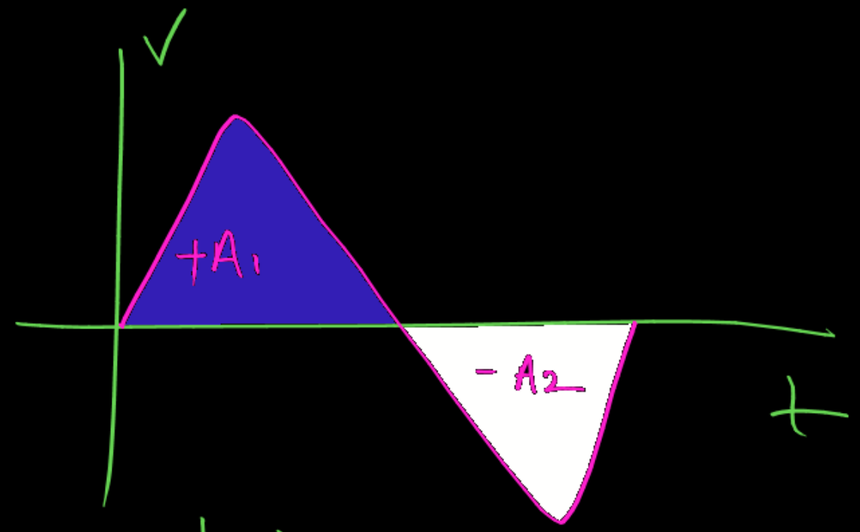
" " " " " " are anti Parallel → t_1 to t_2

(speed) ↓

(14) Velocity - time graph.



$$\text{Area} = \int V\,dt = \text{disp}^{m}$$

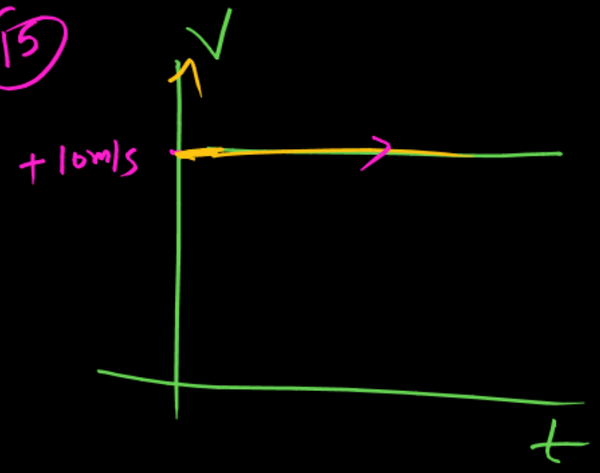


⇒ Velocity → y^{th} co-ordinate V = +ve ↑

Instd. $acc^{n} = \frac{dv}{dt}$ Slope = +ve rot^{n}.

- disp^{m} = $A_1 - A_2$ ✓
- disp^{n} = $A_1 + A_2$ ✓

(15)

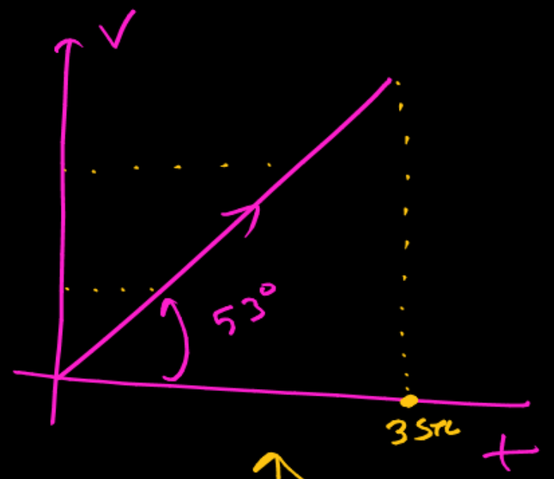


$V = \text{cost}^n = +10\,m/s$

but $a = 0$

disp in 5 sec = ??

(16)



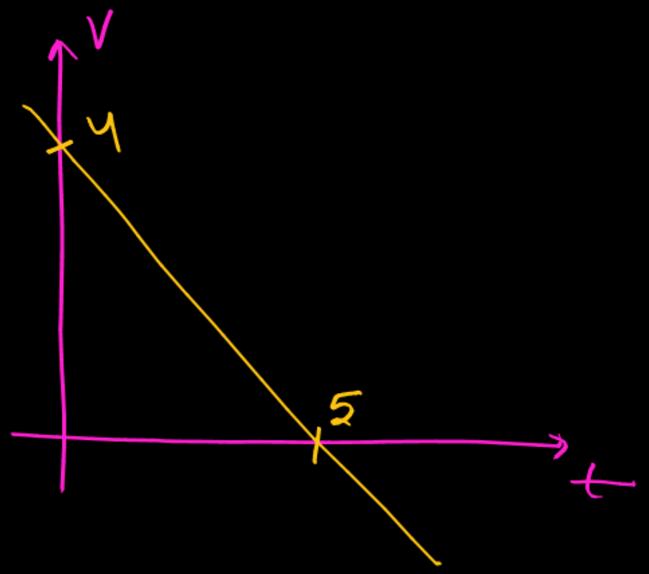
$u = \uparrow$

$a = (+ve)\tan 53^\circ = \text{cot}^n$

$a = +\frac{4}{3}\,m/s = \text{cot}^n$

disp in 3-sec = ??

(17)



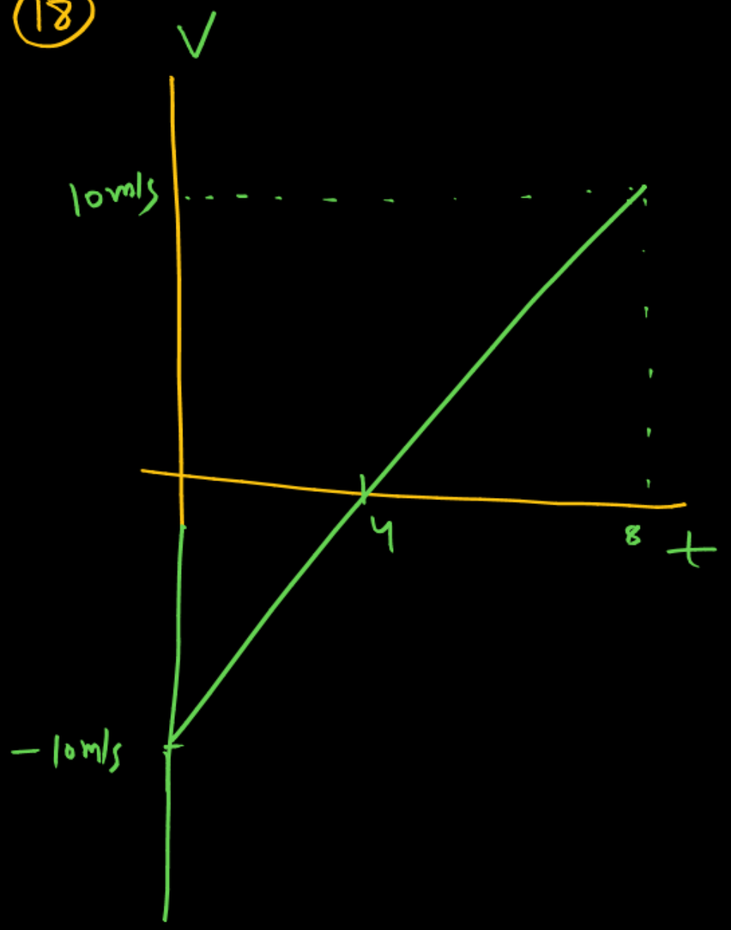
$V = \downarrow$

$a = \text{slope} = -\left(\frac{4}{5}\ m/s^2\right)$

$disp^n = ??$

in 5-sec

(18)



$a =$

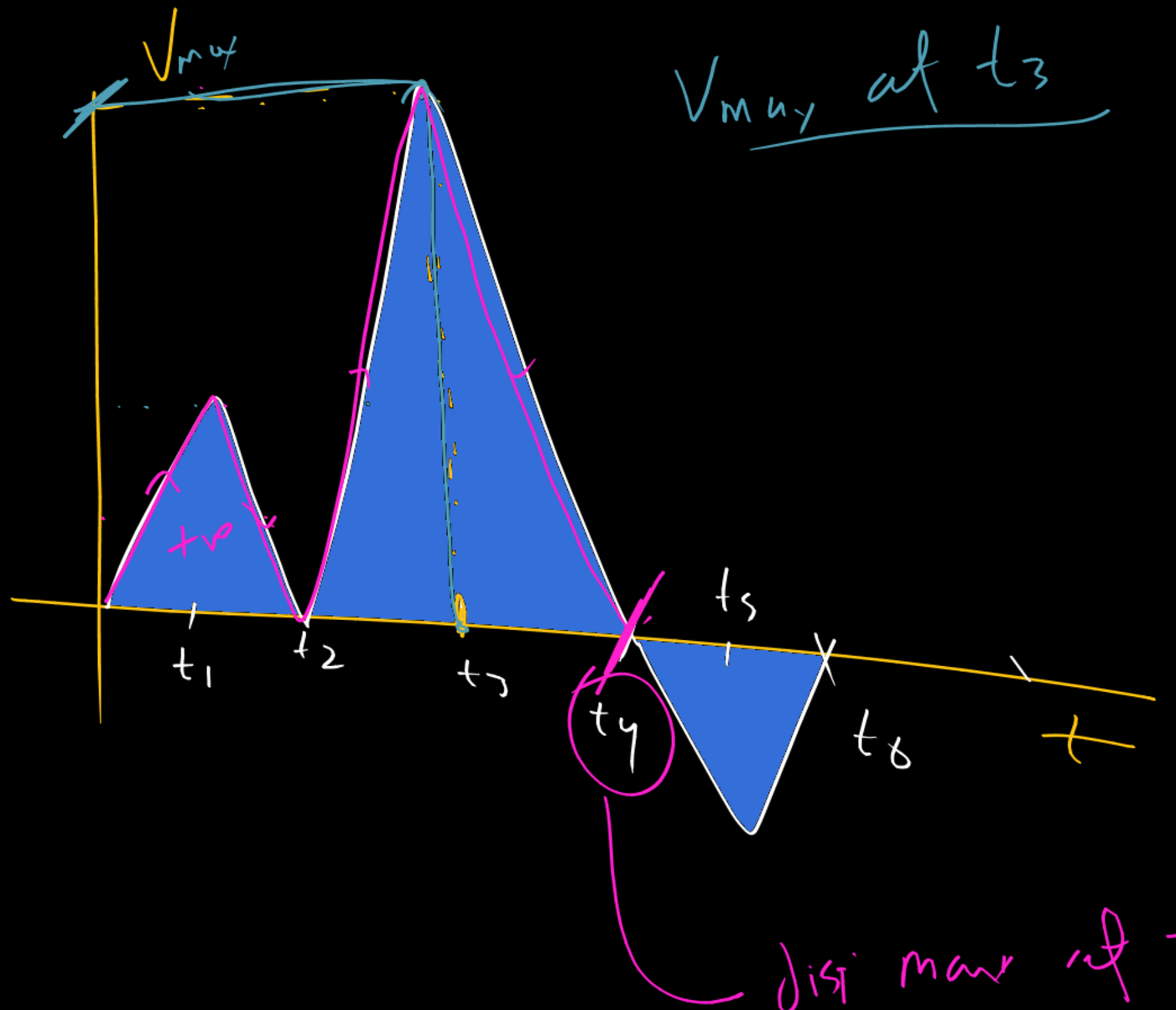
$\text{disp}^{m}_{8\,sec} =$

Avg velocity in 8 sec

dist^{n} in 8-sec

(19) disp^m of Particle max^m at

Area se dekho



(a) t_1

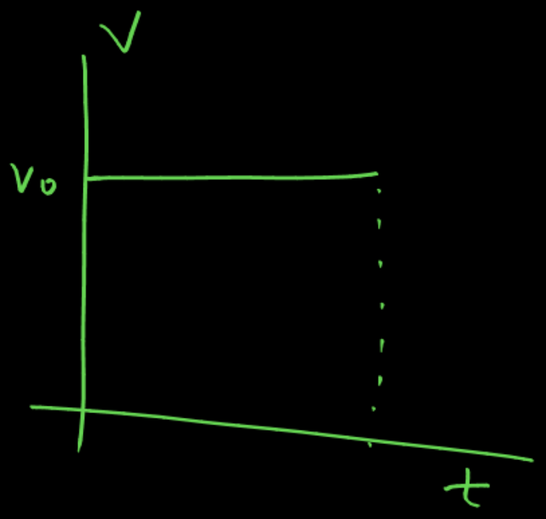
(b) t_2 mR Scam

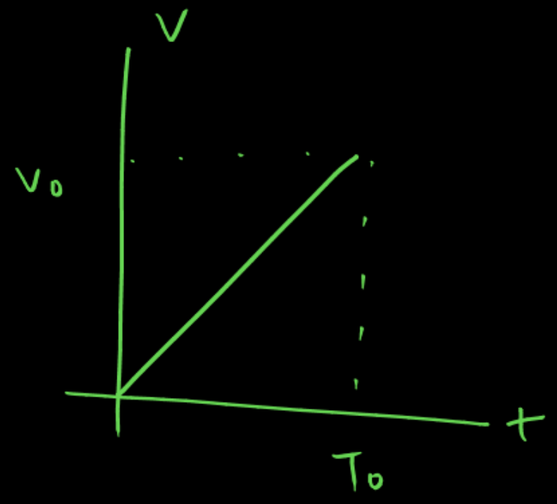
(c) t_3 (−1)

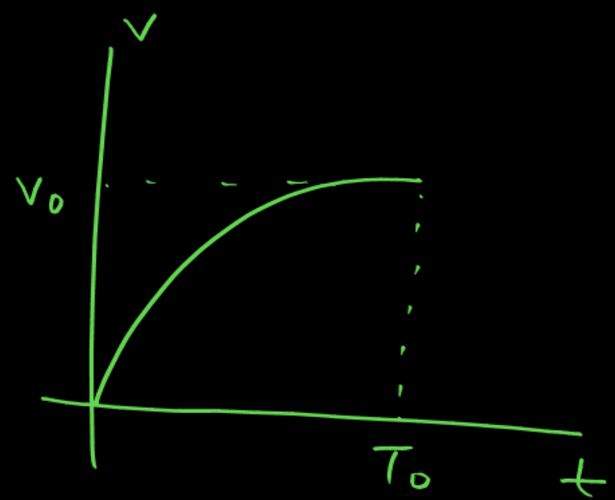
(d) ~~t_4~~ (20) +1

(e) t_6

(20)

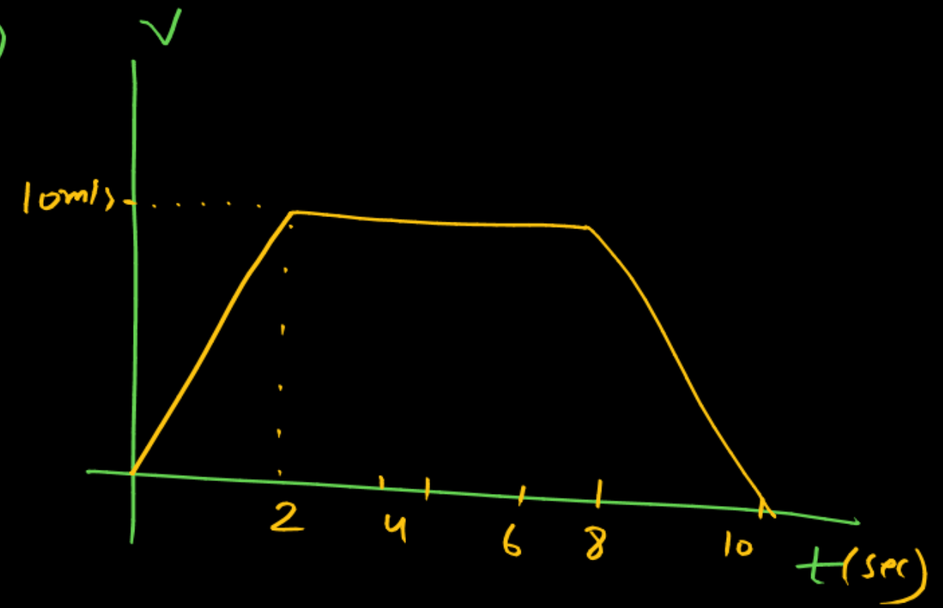






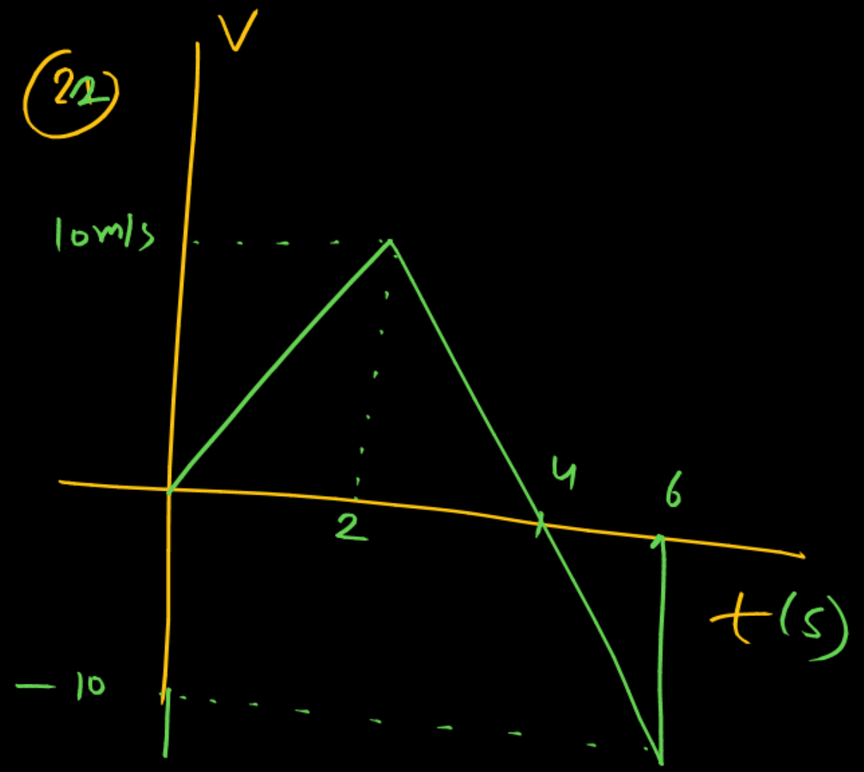
Avg velocity is maxm in which graph $(0-T_0)$

(21)



dist^n & disp^m in (10sec)

(22)



find dist^n & disp^n in 6-sec

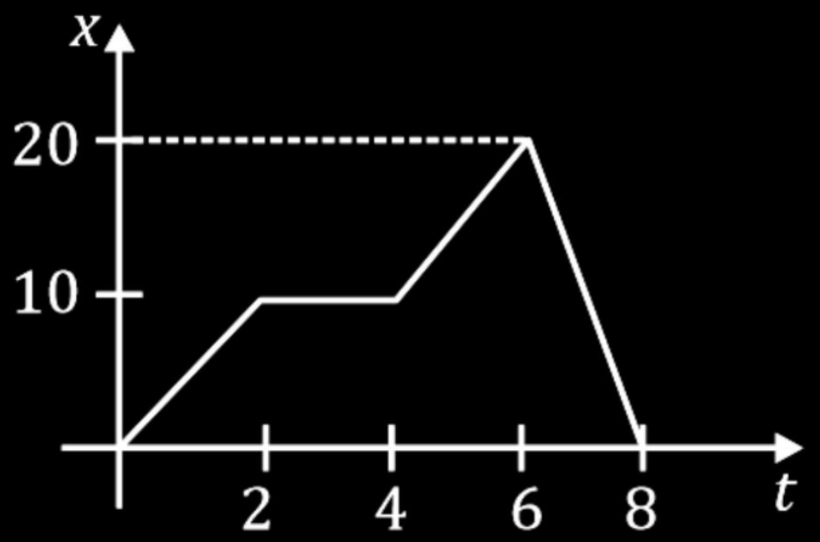
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in noter

Do in rough Not copy

Question

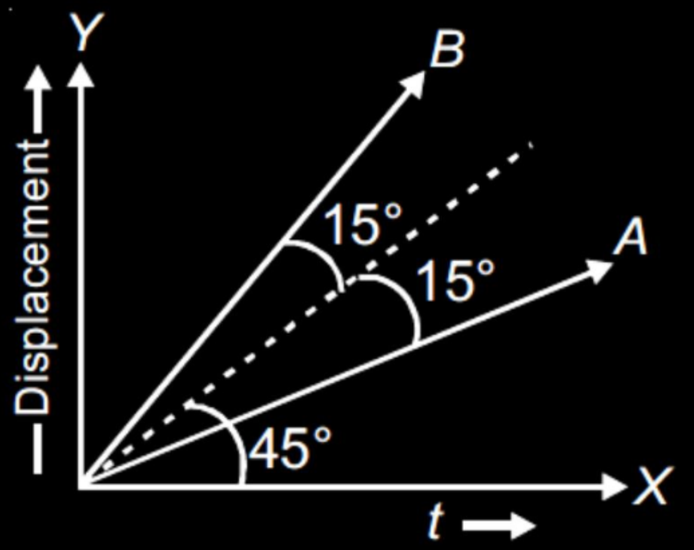
Find avg speed and velocity in 8 sec.



Question

The displacement-time graph for two particles A and B is follows. The ratio v_A/v_B is

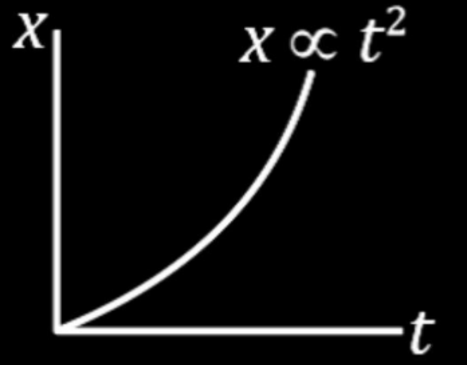
1. 1 : 2
2. $1 : \sqrt{3}$
3. 13 : 1
4. 1 : 3

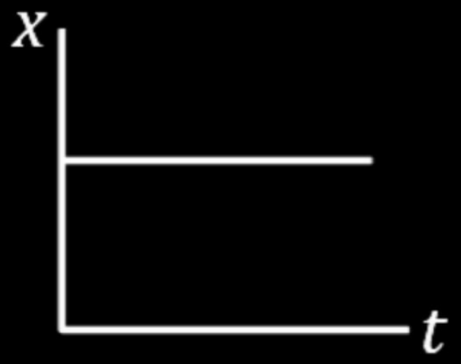


Question

In which graph acceleration is non-zero constant?

1. 

2. 

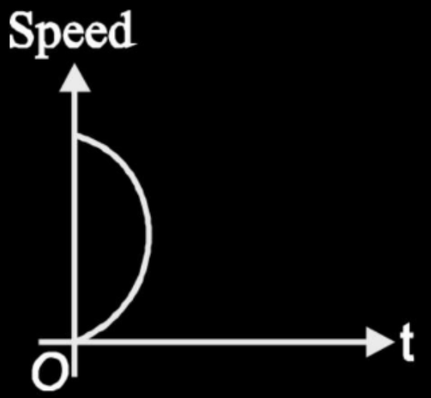
3. 

4. None of these

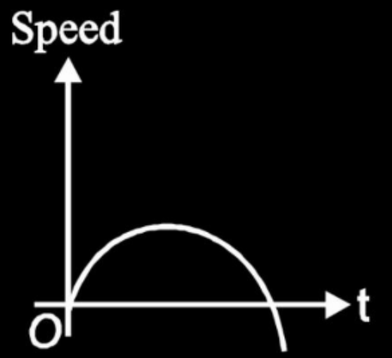
Question

Which one of the following graph for a body moving along a straight line is possible?

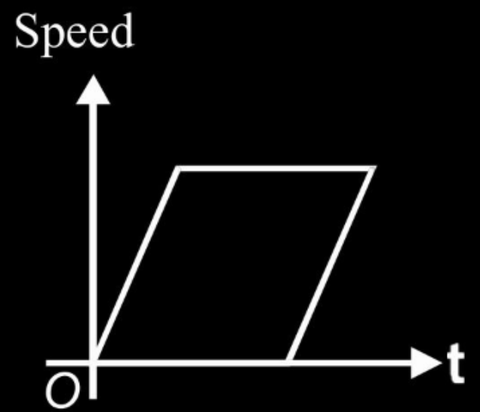
1



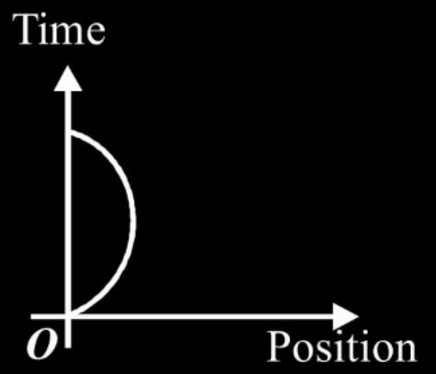
2



3



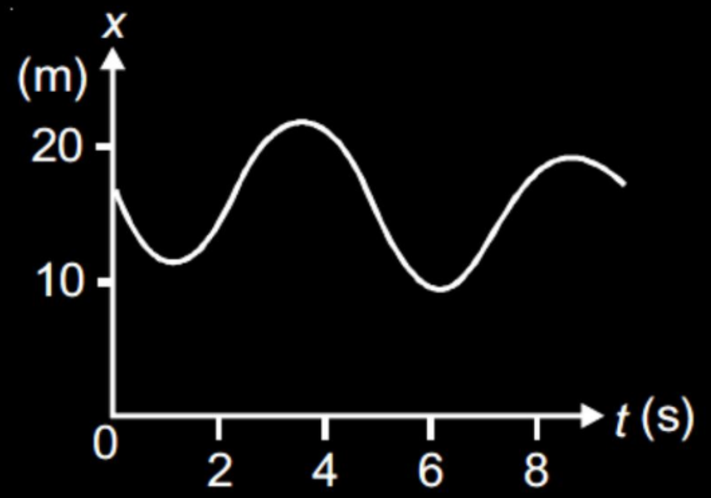
4



Question

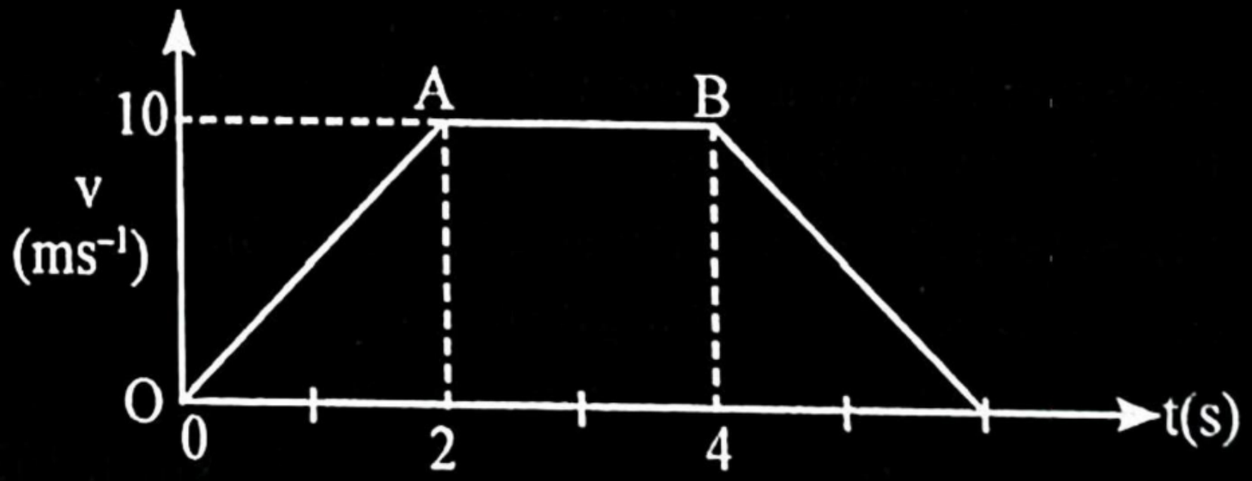
Figure shows the position of a particle moving on the x-axis as a function of time

1. The particle has come to rest 4 times
2. The velocity at $t = 8$ s is negative
3. The velocity remains positive for $t = 2$ s to $t = 6$ s
4. The particle moves with a constant velocity



Question

The velocity-time graph of an object moving along a straight line is shown in figure. What is the distance covered by the object between $t = 0$ to $t = 4s$? **[JEE Main 2025]**



1. 10 m
2. 30 m
3. 13 m
4. 11 m

Question

A particle starts from rest, accelerates at 2 m/s^2 for 10s and then goes with constant speed for 30s and then decelerates at 4 m/s^2 till it stops. What is the distance travelled by it

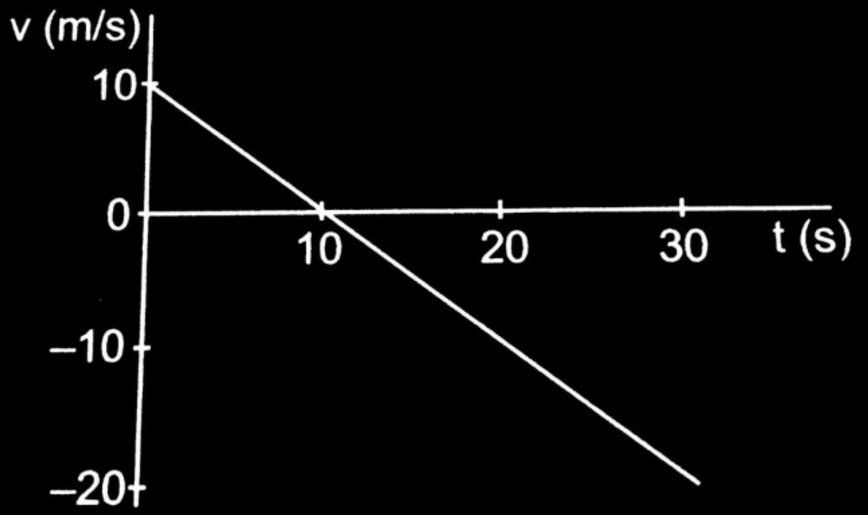
1. 750 m
2. 800 m
3. 700 m
4. 850 m

Question

A particle starts from rest and moves with constant acceleration 4 m/s^2 for 10 sec after that it moves with constant velocity for 20 more sec then it retards with 8 m/s^2 and stop. Find total distance moved by object.

Question

The velocity-time plot for a particle moving on a straight line shown in the figure



1. The particle has a constant acceleration
2. The particle has never turned around
3. The particle has a zero displacement
4. The average speed in the interval 0 to 10s is the same as the average speed in the interval 10s to 30s

Question

Object starts his motion from rest and acceleration = 5 m/s^2 then find displacement in 4 sec and in 4th sec.

Thank

You